

Homework 2: Nico Boyle, Owen Moorhus, Asher Olgeirson

1. Let k be an integer.

Claim: If $5k + 3$ is even, then k is odd.

Prove the claim by first converting to contrapositive.

Proof :

By contrapositive, assume k is an even integer

$k = 2n$ where n is an integer

$$5k + 3 = 5(2n) + 3$$

$$= 10n + 3$$

$$= 10n + 2 + 1$$

$$= 2(5n + 1) + 1$$

$$= (2j) + 1 \text{ where } j \text{ is an integer}$$

$$= 5k + 3 = 2j + 1 \text{ which is odd by definition, thus proving } k \text{ is odd by contrapositive}$$



2. Suppose that x and y are positive real numbers. Prove by contradiction that (for all such x, y)
 $x + y \geq 2\sqrt{xy}$

Proof :

by contradiction, assume $x + y < 2\sqrt{xy}$

$$x^2 + 2xy + y^2 < 4xy$$

$$x^2 - 2xy + y^2 < 0$$

$$(x - y)^2 < 0$$

This is a contradiction since the square of a real number is always non-negative.



3. Prove the following, by smallest counterexample.

$$\text{For all } n \geq 1, \sum_{k=1}^n k(k+1) = \frac{n(n+1)(n+2)}{3}$$

Proof :

$$\text{True Statement: } 2 + 6 + 12 + \dots + n(n-1) = \frac{n(n-1)(n+1)}{3}$$

$$\text{Smallest Counterexample: } 2 + 6 + 12 + \dots + n(n-1) + n(n+1) \neq \frac{n(n+1)(n+2)}{3}$$

$$\frac{n(n-1)(n+1)}{3} + n(n+1) \neq \frac{n(n+1)(n+2)}{3}$$

$$n(n-1)(n+1) + 3n(n+1) \neq n(n+1)(n+2)$$

$$n(n+1) \cdot (n-1+3) \neq n(n+1)(n+2)$$

$$n(n+1) \cdot (n+2) \neq n(n+1)(n+2)$$

$$n(n+1)(n+2) \neq n(n+1)(n+2)$$

This is a contradiction. Thus, there is no smallest counter example and the claim is true.

